Elementary Problems
Existence Problem
Solutions

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Basic Work

1. We provide one possible configuration. Suppose the numbers assigned to the vertices are

$$\underbrace{11\cdots1}_{n\text{ times}},\ \underbrace{11\cdots1}_{n-1\text{ times}}2,\ \underbrace{11\cdots1}_{n-2\text{ times}}22,\ \underbrace{11\cdots1}_{n-2\text{ times}}21,\ \underbrace{11\cdots1}_{n-3\text{ times}}221,\ \underbrace{11\cdots1}_{n-3\text{ times}}211,$$
$$\ldots,\ 22\underbrace{11\cdots1}_{n-2\text{ times}},\ 2\underbrace{11\cdots1}_{n-1\text{ times}}$$

   in order. One checks that all conditions are satisfied.

2. Yes, such subsets exist. For example, consider $A_j = \{2^{j-1}\}$ for $j = 1, 2, \ldots, n-1$ and $A_n = \{k \cdot 2^{n-1} : k \in \mathbb{N}\}$. For any positive integer m, any representation of m as a sum of integers belonging to different subsets has the form

$$m = a_1 + a_2 \cdot 2 + a_3 \cdot 2^2 + \cdots + a_{n-1} \cdot 2^{n-2} + a_n \cdot 2^{n-1}$$

   where each a_j equals 0 or 1 for $j = 1, 2, \ldots, n-1$, and a_n is a nonnegative integer. Using the binary representation of m, we can always find such a representation, and the representation is unique. So the condition is satisfied.

3. Let the n-gon be $A_1A_2\cdots A_{2n+1}$ and let B_j be the midpoint of A_j and A_{j+1} for each $j = 1, 2, \ldots, n$ (where $A_{2n+2} = A_1$). Assign the number $2n + 3 - 2k$ to A_{2k-1} for $k = 1, 2, \ldots, n+1$, the number $4n + 3 - 2k$ to A_{2k} for $k = 1, 2, \ldots, n$, and the number $2k$ to B_k for $k = 1, 2, \ldots, 2n+1$. Then the numbers $A_1, B_1, A_2, B_2, \ldots, A_{2n+1}, B_{2n+1}$ are

$$2n+1, 2, 4n+1, 4, 2n-1, 6, 4n-1, 8, \ldots, 3, 4n-2, 2n+3, 4n, 1, 4n+2$$

   in that order.

   Clearly, all numbers $1, 2, \ldots, 4n+2$ are used once. The sum of numbers on the side formed by $A_{2k-1}, B_{2k-1}, A_{2k}$ (for $1 \leqslant k \leqslant n$) is

$$(2n + 3 - 2k) + 2(2k - 1) + (4n + 3 - 2k) = 6n + 4,$$

   while the sum of numbers on the side formed by A_{2k}, B_{2k}, A_{2k+1} (for $1 \leqslant k \leqslant n$) is

$$(4n + 3 - 2k) + 2(2k) + (2n + 1 - 2k) = 6n + 4.$$

   Also, the sum of numbers on the side formed by A_{2n+1}, B_{2n+1}, A_1 is

$$(1) + 2(2n + 1) + (2n + 1) = 6n + 4.$$

   Hence, all sums are equal.

4. The zeros can be partitioned into some blocks of consecutive zeros. If there is a block with at least 3 zeros, then we are done. So we may assume each block has 1 or 2 zeros. Thus, there are at least $\left\lceil \frac{2n+1}{2} \right\rceil = n+1$ blocks of zeros. These blocks are separated by some ones.

   Since there are only $2n+1$ ones, there are two blocks separated by at most $\left\lfloor \frac{2n+1}{n+1} \right\rfloor = 1$ one. Then the neighbours of this one are both zeros as desired.

5. Consider arbitrary integers $a, b, c > N$ which are of the same colour, say black. If one of a, b, c is the sum of the other two numbers, then we are done. If one of $a+b$, $b+c$, $c+a$ is black, then we are done. Otherwise, we may assume $a+b$, $b+c$, $c+a$ are white. For similar reason, we may assume each of

$$\begin{aligned}(a+b)+(a+c) &= 2a+b+c,\\ (a+b)+(b+c) &= a+2b+c,\\ (a+c)+(b+c) &= a+b+2c\end{aligned}$$

   is black.

   If $a+b+c$ is black, then we are done since $a+(a+b+c) = 2a+b+c$ and each of a, $a+b+c$ and $2a+b+c$ is black. If $a+b+c$ is white, then we further consider $2a+2b+c$.

   Firstly, if $2a+2b+c$ is black, then the numbers b, $2a+b+c$ and $2a+2b+c$ are black with $b+(2a+b+c) = 2a+2b+c$. If $2a+2b+c$ is white, then the numbers $a+b$, $a+b+c$ and $2a+2b+c$ are white with $(a+b)+(a+b+c) = 2a+2b+c$.

   Therefore, in any case, we can find $m, n > N$ such that the integers $m, n, m+n$ are of the same colour.

6. Let the vertices be $V_1, V_2, \ldots, V_{100}$ and let the two numbers on V_j be a_j and b_j.

   Firstly, suppose there exist two adjacent vertices such that a number on one vertex is different from both numbers on another vertex. WLOG assume $a_1 \neq a_2$ and $a_1 \neq b_2$. We remove b_1. As a_{100} and b_{100} are distinct, we can remove one of them so that the remaining number is distinct from a_1. WLOG assume $a_{100} \neq a_1$ and we remove b_{100}. Inductively, we may remove a number from the vertices $V_{99}, V_{98}, \ldots, V_2$, say $b_{99}, b_{98}, \ldots, b_2$, such that the remaining numbers satisfy $a_{k+1} \neq a_k$ for $k = 99, 98, \ldots, 2$. As $a_2 \neq a_1$, any two adjacent numbers in the cycle $a_1, a_2, \ldots, a_{100}$ are distinct.

   Now, suppose for any two adjacent vertices, both numbers on one vertex are the same as the numbers on another vertex. This means the two numbers on all vertices are exactly the same. Thus, we may assume the two numbers on each vertex are a and b. Then we simply remove a for $V_1, V_3, \ldots, V_{99}$ and remove b for $V_2, V_4, \ldots, V_{100}$.

7. Yes. For each n, let a_n be the number of primes among $n, n+1, \ldots, n+2003$. Consider a_n and a_{n+1}. Among the consecutive integers involved, there are 2003 terms in common

and only 1 term is different. Therefore, a_{n+1} is equal to one of

$$a_n - 1,\ a_n,\ a_n + 1.$$

It is obvious that $a_1 > 12$ as $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 33, 37$ are primes.

From the given fact, we have $a_k = 0$ when $k = 2005! + 2$. In order that the value of a_n decreases from a_1 to 0, there must be some intermediate term for which $a_n = 12$. This is what we need.

8. We label the n people from 0 to $n-1$ in the clockwise direction initially. After the break, we may assume person 0 does not change the seat. For $1 \leqslant k \leqslant n-1$, suppose the person k has moved a_k steps clockwise. This means the person has moved to the position $k + a_k$ or $k + a_k - n$.

   If $a_k = 0$ for some k, then both person 0 and person k have not moved, and they are the pair we want.

   If $a_j = a_k$ for some $j \neq k$, then both person j and person k have moved exactly a_j steps clockwise. Hence, they are the pair we want.

   Now, we assume $a_1, a_2, \ldots, a_{n-1}$ are distinct nonzero integers. As $1 \leqslant a_j \leqslant n-1$, they must be a permutation of $1, 2, \ldots, n-1$. As the sum of the positions that all people are sitting is $0 + 1 + \cdots + n$, we have

$$0 + (1 + a_1) + (2 + a_2) + \cdots + (n - 1 + a_{n-1}) \equiv 0 + 1 + \cdots + n \pmod{n}.$$

   This can be simplified to $a_1 + a_2 + \cdots + a_{n-1} \equiv 0 \pmod{n}$. However, as $a_1, a_2, \ldots, a_{n-1}$ is a permutation of $1, 2, \ldots, n-1$, we have

$$a_1 + a_2 + \cdots + a_{n-1} \equiv 1 + 2 + \cdots + (n-1) = \frac{n(n-1)}{2}.$$

   This cannot be a multiple of n since $\dfrac{n-1}{2}$ is not an integer. This is a contradiction. Therefore, we can always find such a pair of persons.

9. Yes. The following gives an example.

7	13	19	34	40	46	61	67	73
35	41	47	62	68	74	8	14	20
63	69	75	9	15	21	36	42	48
1	16	22	28	43	49	55	70	76
29	44	50	56	71	77	2	17	23
57	72	78	3	18	24	30	45	51
4	10	25	31	37	52	58	64	79
32	38	53	59	65	80	5	11	26
60	66	81	6	12	27	33	39	54

This is constructed as follows. We first consider two boards A and B:

1	2	3	4	5	6	7	8	9
4	5	6	7	8	9	1	2	3
7	8	9	1	2	3	4	5	6
1	2	3	4	5	6	7	8	9
4	5	6	7	8	9	1	2	3
7	8	9	1	2	3	4	5	6
1	2	3	4	5	6	7	8	9
4	5	6	7	8	9	1	2	3
7	8	9	1	2	3	4	5	6

7	4	1	7	4	1	7	4	1
8	5	2	8	5	2	8	5	2
9	6	3	9	6	3	9	6	3
1	7	4	1	7	4	1	7	4
2	8	5	2	8	5	2	8	5
3	9	6	3	9	6	3	9	6
4	1	7	4	1	7	4	1	7
5	2	8	5	2	8	5	2	8
6	3	9	6	3	9	6	3	9

Each entry in the desired board is 9 times the corresponding entry in A plus the corresponding entry in B minus 9. Each digit in A appears 9 times. The 9 corresponding entries in B are all distinct. So all numbers of the form $9a + b - 9$ are formed, where $1 \leqslant a, b \leqslant 9$. That means the numbers are exactly 1 to 81. Since every 3×3 square in both A and B consists of numbers from 1 to 9, the sum of numbers in each 3×3 square in the desired board is

$$9(1 + 2 + \cdots + 9) + (1 + 2 + \cdots + 9) - 9 \times 9 = 369.$$

Pigeonhole Principle

10. Consider the n pairs
$$(1, 2), (3, 4), \ldots, (2n - 1, 2n).$$
Since $|S| = n + 1$, there are two elements in S belonging to the same pair by the pigeonhole principle. The two integers in the same pair are consecutive, and hence they are relatively prime.

11. We partition the 64 cells of the chessboard into 8 groups. The cell (i, j) belongs to the kth group if $i + j \equiv k \pmod 8$. By the pigeonhole principle, there exists a group containing at least $\left\lceil \dfrac{17}{8} \right\rceil = 3$ rooks. These 3 rooks do not threaten each other. Indeed, if a rook in the cell (i_1, j_1) and another in the cell (i_2, j_2) threaten each other, then either $i_1 = i_2$ or $j_1 = j_2$. In both cases, since $i_1 + j_1 \equiv i_2 + j_2 \pmod 8$, we must have $(i_1, j_1) = (i_2, j_2)$. This is a contradiction. So we are done.

12. Let the positive integers be $a_1 < a_2 < \cdots < a_{20}$. Consider the 19 differences
$$a_2 - a_1,\ a_3 - a_2,\ \ldots,\ a_{20} - a_{19}.$$

Each of them is a positive integer. If no 4 of them are equal, then the sum of these differences is at least

$$1+1+1+2+2+2+\cdots+6+6+6+7=70.$$

However, we have

$$(a_2-a_1)+(a_3-a_2)+\cdots+(a_{20}-a_{19})=a_{20}-a_1 \leqslant 70-1=69.$$

This is a contradiction. Therefore, there are 4 equal differences.

13. For $j=1,2,\ldots,77$, let a_j be the total number of games played from day 1 to day j inclusive. Note that

$$1 \leqslant a_1 < a_2 < \cdots < a_{77} \leqslant 132$$

and

$$22 \leqslant a_1+21 < a_2+21 < \cdots < a_{77}+21 \leqslant 153.$$

Among the 154 numbers

$$a_1, a_2, \ldots, a_{77},\ a_1+21,\ a_2+21,\ \ldots,\ a_{77}+21,$$

there are two equal numbers by the pigeonhole principle since they lie between 1 and 153. As all a_j's are distinct, we must have $a_j = a_i + 21$ for some indices $i<j$. Hence, exactly 21 games are player from day $i+1$ to day j inclusive.

14. Let the sequence be $a_1, a_2, \ldots, a_t$ with $t \geqslant (m-1)(n-1)+1$. For the term a_j, define I_j to be the maximum number of terms of a monotonic increasing subsequence using a_j as the initial term, and define D_j analogously to be the maximum number of terms of a monotonic decreasing subsequence using a_j as the initial term.

We claim that for any $i \neq j$, either $I_i \neq I_j$ or $D_i \neq D_j$. Suppose on the contrary that $I_i = I_j$ and $D_i = D_j$ for some $i<j$. WLOG assume $a_i \leqslant a_j$. Then we can find a monotonic increasing subsequence of length I_j with a_j as the initial term. We can put the extra term a_i before the term a_j to form another monotonic increasing subsequence of length $I_j+1 = I_i+1$. This contradicts the definition of I_i. Therefore, all the pairs (I_j, D_j) are distinct.

Now, suppose on the contrary that the longest monotonic increasing subsequence has length $m-1$ and the longest monotonic decreasing subsequence has length $n-1$. Then $I_j \leqslant m-1$ and $D_j \leqslant n-1$ for each j. There are only $(m-1)(n-1)$ possibilities of the pair (I_j, D_j). By the pigeonhole principle, as $t > (m-1)(n-1)$, there exist distinct indices i and j with $(I_i, D_i) = (I_j, D_j)$. This contradicts to the above observation.

Therefore, there exists a monotonic increasing subsequence of length m or a monotonic decreasing subsequence of length n.

15. Consider the integers

$$a_1^2,\ a_1^2 + a_2^2,\ \ldots,\ a_1^2 + a_2^2 + \cdots + a_n^2.$$

If any of these is divisible by n, then we are done. Otherwise, these n integers can only be congruent to $1, 2, \ldots, n-1$ modulo n. By the pigeonhole principle, there are two of them which are congruent modulo n. Suppose

$$a_1^2 + a_2^2 + \cdots + a_i^2 \equiv a_1^2 + a_2^2 + \cdots + a_j^2 \pmod{n}$$

for some $i < j$. Then

$$a_{i+1}^2 + a_{i+2}^2 + \cdots + a_j^2 \equiv 0 \pmod{n}.$$

Thus, we can choose the subsequence $a_{i+1}, a_{i+2}, \ldots, a_j$.

In any case, such a subsequence exists.

16. Let the numbers be $a_1, a_2, \ldots, a_{99}$. Consider the numbers

$$a_1,\ a_1 + a_2,\ \ldots,\ a_1 + a_2 + \cdots + a_{99}. \tag{1}$$

These numbers are not multiples of 100. If two of them are congruent modulo 100, say

$$a_1 + \cdots + a_i \equiv a_1 + \cdots + a_j \pmod{100}$$

for some $i < j$, then

$$a_{i+1} + a_{i+2} + \cdots + a_j \equiv 0 \pmod{100}.$$

This contradicts the given condition. Thus, the 99 numbers in (1) are pairwise incongruent modulo 100. Hence, they must cover all residue classes from 1 to 99 modulo 100.

Note that $a_2 \not\equiv 0 \pmod{100}$. This number must be congruent to one of the numbers in (1). If

$$a_2 \equiv a_1 + a_2 + \cdots + a_j \pmod{100}$$

for some $j \geqslant 2$, then

$$a_1 + a_3 + a_4 + \cdots + a_j \equiv 0 \pmod{100}.$$

This is a contradiction. Thus, the only possibility is $a_1 \equiv a_2 \pmod{100}$. From the given range, this implies $a_1 = a_2$. Due to symmetry, all the cards contain the same number.

17. Consider any sequence (a_m) in B. For any sequence (b_m) in B (which can be (a_m)), there exists a unique sequence (c_m) such that the product of (a_m) and (c_m) is (b_m). Indeed, we need $a_j c_j = b_j$ for each j, which is the same as $c_j = a_j b_j$ as each entry is ± 1. This uniquely determines (c_m). Hence, there are exactly k sequences (c_m) such that the product of (a_m) and (c_m) belongs to B. Thus, each sequence in B contributes

k to the total number of elements in all the intersections when (c_m) runs through all possibilities.

As there are k sequences in B, they contribute a total of k^2 to the total number of elements in all the intersections. By the pigeonhole principle, there is a sequence (c_m) such that the corresponding intersection has at most $\dfrac{k^2}{2^n}$ sequences.

18. We first consider the special case $n = 2^k$ for $k \geqslant 1$. We shall prove by induction on k that there exists a multiple of 2^k consisting of exactly k nonzero digits. Indeed, this clearly holds when $k = 1$. Assume 2^m has such a multiple a. Consider the numbers $a + 10^m$ and $a + 2 \cdot 10^m$. As both of them are multiples of 2^m and their difference 10^m is not a multiple of 2^{m+1}, exactly one of them is a multiple of 2^{m+1}. Also, this number contains exactly $m + 1$ nonzero digits.

    A similar argument applies to the case $n = 5^k$ for $k \geqslant 1$. Indeed, we simply need to consider the numbers

    $$a + 10^m,\ a + 2 \cdot 10^m,\ a + 3 \cdot 10^m,\ a + 4 \cdot 10^m,\ a + 5 \cdot 10^m$$

    in the lifting process.

    For the general case, let $n = 2^\alpha 5^\beta n'$ where $(n', 10) = 1$. As one of α, β is zero, the above shows that there exists a multiple a of $2^\alpha 5^\beta$ consisting of nonzero digits only. Consider the numbers

    $$a, \overline{aa}, \overline{aaa}, \ldots$$

    obtained by putting more copies of a together. By the pigeonhole principle, two of these numbers are congruent modulo n'. Suppose

    $$\underbrace{\overline{aa\cdots a}}_{i \text{ times}} \equiv \underbrace{\overline{aa\cdots a}}_{j \text{ times}} \pmod{n'}$$

    for some $i < j$. Then this implies

    $$\overline{\underbrace{aa\cdots a}_{j-i \text{ times}}00\cdots 0}$$

    is a multiple of n'. As $(n', 10) = 1$, we can remove the zeros at the end to obtain a multiple $\overline{aa\cdots a}$ of n' consisting of nonzero digits. Note that this is also a multiple of $2^\alpha 5^\beta$. Hence, it is divisible by n.

19. There are $2^{10} - 1 = 1023$ nonempty subsets of a set of 10 numbers. Note that the sum of numbers in a subset is at most $90 + 91 + \cdots + 99 < 1000$. Thus, there are less than 1000 possible sums. By the pigeonhole principle, two of the subsets must have the same sum of numbers. By removing the common elements in the two subsets, they become disjoint subsets and the sums remain the same. These are the subsets we need.

20. We consider all the 1000-element subsets of $\{n_1, n_2, \ldots, n_{2000}\}$. There are $\binom{2000}{1000}$ such subsets. For each such subset S, we let a_S and b_S be the sum of elements and sum of squares of elements of S respectively.

Trivially, we have

$$a_S < 10^{100} \cdot 1000 = 10^{103}$$

and

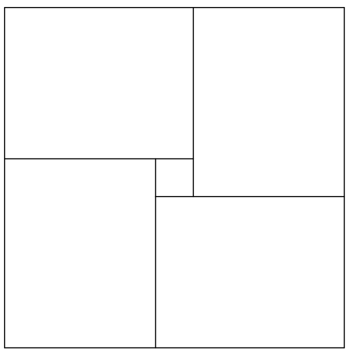
$$b_S < 10^{200} \cdot 1000 = 10^{203}.$$

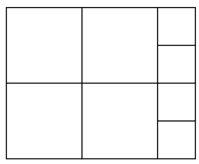
Hence, there are at most $10^{103} \cdot 10^{203} = 10^{306}$ possible pairs (a_S, b_S). Note that

$$\begin{aligned}\binom{2000}{1000} &= \left(\frac{2000}{1000} \times \frac{1999}{999} \times \cdots \times \frac{1004}{4}\right) \times \frac{1003}{3} \times \frac{1002}{2} \times \frac{1001}{1} \\ &> 2^{997} \cdot 300 \cdot 500 \cdot 1000 \\ &= (2^{10})^{99} \cdot 2^7 \cdot 15 \cdot 10^7 \\ &> (10^3)^{99} \cdot 10^{10} > 10^{306}.\end{aligned}$$

By the pigeonhole principle, two of these subsets correspond to the same pair (a_S, b_S). This means the two subsets have the same sum of elements and the same sum of squares of elements. By deleting the same elements in both subsets, they become disjoint and they still have the same number of elements, the same sum of elements and the same sum of squares of elements. Also, as they are distinct, one of them, and hence both of them, is nonempty.

21. Suppose on the contrary that each 2×2 block contains at most 2 red squares. Partition the chessboard into four copies of 4×5 sub-board together with a unit square as shown. By the pigeonhole principle, one of the sub-board contains at least $\left\lceil \frac{46-1}{4} \right\rceil = 12$ red squares. WLOG assume the top left sub-board is such a board.





We further partition this 4×5 board into four copies of 2×2 blocks together with four unit squares. Then each 2×2 block has at most 2 red squares. Since $12 = 2 \times 4 + 4$, all of the four unit squares are red. By symmetry, the four squares on the other short edge are also red. Then the four squares in the middle column are red. In other words, this is the unique case for which this board may contain 12 red squares.

In that case, the first row of the bottom left 5×4 board contains at most 2 red squares. So this board contains at most $2 + 4 \times 2 = 10$ red squares by considering the same partition as above. Thus, the other two 4×5 boards contain at least $46 - 1 - 12 - 10 = 23$ red squares in total. By the pigeonhole principle, one of them contains at least $\left\lceil \frac{23}{2} \right\rceil = 12$ red squares. If it is the bottom right board, then these two boards contain at most $12 + 10 = 22$ red squares as above, which is a contradiction. If it is the top right board, then the top left board contains at most 10 red squares as above, which is also a contradiction.

Hence, there must be a 2×2 block having at least three red squares.

22. Let m be a sufficient large integer. Consider a regular polygon with m sides lying on the circle. By the pigeonhole principle, there exist $\left\lceil \frac{m}{N} \right\rceil$ vertices having the same colour. In addition, we can choose at least half of them lying on the same side of a diameter of the circle. Removing one vertex if necessary, we may assume there are $2k$ vertices on the same semicircle having the same colour where $2(2k+1) \geqslant \left\lceil \frac{m}{N} \right\rceil$. This implies

$$k \geqslant \frac{1}{2}\left(\frac{m}{2N} - 1\right) > \frac{1}{2} \cdot \frac{m}{4N} = \frac{m}{8N}$$

as m is large compared to N (indeed we only need $m > 4N$).

Let the $2k$ vertices be $V_1, V_2, \ldots, V_{2k}$. Consider the segments V_iV_j where i is odd, j is even and $i < j$. There are

$$k + (k-1) + \cdots + 1 = \frac{k(k+1)}{2}$$

such segments. Since each segment is a side or a diagonal of the regular m-gon, its length only has $\left\lfloor \frac{m}{2} \right\rfloor$ possibilities. By the pigeonhole principle, as

$$\frac{k(k+1)}{2} > \frac{k^2}{2} > \frac{1}{2}\left(\frac{m}{8N}\right)^2 = \frac{m^2}{128N^2} > \left\lfloor \frac{m}{2} \right\rfloor$$

for sufficiently large m, say $m > 64N^2$, there must be 2 different segments V_iV_j and V_sV_t having the same length. Note that i and s are odd, j and t are even, $i < j$ and $s < t$. If $i = s$, then $j = t$ since $V_iV_j = V_sV_t$ and the vertices lie on the same semicircle. Similarly, we cannot have $j = t$. This shows V_i, V_j, V_s, V_t are pairwise distinct. Then $V_iV_jV_sV_t$ is an isosceles trapezoid if $j < s$, and $V_iV_sV_jV_t$ is an isosceles trapezoid if $s < j$. Its vertices have the same colour and so we are done.

23. There are $\binom{8}{2} = 28$ possible pairs of x-coordinates which are the two vertical boundaries of a rectangle. Consider those rectangles with vertical boundaries $x = x_1$ and $x = x_2$. If there are 8 of them, since the lower horizontal boundaries of them can only be $y = 0$, $y = 1$, …, $y = 6$, there are 2 of them having the same lower horizontal boundary by the pigeonhole principle. Since they have 3 boundaries in common, one of them must lie inside the other.

    As $196 = 28 \times 7$, if the above case does not hold, then for each pair of vertical boundaries, there are 7 rectangles having these as boundaries. Also, in order that the above argument does not work, we may assume their lower horizontal boundaries are $y = 0$, $y = 1$, …, $y = 6$, while their upper horizontal boundaries are $y = 1$, $y = 2$, …, $y = 7$ in some order. Since the upper horizontal boundary must lie above the lower horizontal boundary, the rectangle with lower boundary $y = a$ must have upper boundary $y = a + 1$ for each a.

    The remaining case is uniquely determined. In particular, note that we have the two rectangles $\mathcal{R}_1$ and $\mathcal{R}_2$ where $\mathcal{R}_1$ is bounded by $x = 0$, $x = 1$, $y = 0$, $y = 1$, and $\mathcal{R}_2$ is bounded by $x = 0$, $x = 2$, $y = 0$, $y = 1$. Then $\mathcal{R}_1$ is contained in $\mathcal{R}_2$. So we are done in every case.

24. Let the integer k appears in r_k different rows and c_k different columns. Then they must lie in those cells which are the intersection of these r_k rows and c_k columns. There are $r_k c_k$ intersecting cells, and hence $r_k c_k \geqslant 17$. By the AM-GM inequality, we have

    $$r_k + c_k \geqslant 2\sqrt{r_k c_k} = 2\sqrt{17} > 8.$$

    This implies $r_k + c_k \geqslant 9$ as the variables are integers. Summing over all k's, we get

    $$r_1 + r_2 + \cdots + r_{17} + c_1 + c_2 + \cdots + c_{17} \geqslant 153.$$

    The left-hand side counts the number of pairs (k, L) for which the integer k lies in the row or column L as there are $r_k + c_k$ pairs whose first entries are k. Since there are 34 choices for L, by the pigeonhole principle, there exists a row or column which is the second entry of at least $\left\lceil \frac{153}{34} \right\rceil = 5$ pairs. This means it contains at least 5 different integers as desired.

25. For convenience, we use numbers on the real line to denote the time when the mathematicians were asleep. This means each of the 5 mathematicians was asleep in two closed intervals. Let the 10 intervals be

    $$[a_1, b_1], [a_2, b_2], \ldots, [a_{10}, b_{10}]$$

    respectively so that $a_1 \leqslant a_2 \leqslant \cdots \leqslant a_{10}$.

    Next, for each of the $\binom{5}{2} = 10$ pairs of mathematicians, they were sleeping simultaneously at some moment. Each such pair corresponds to an interval which is the first time when they fell asleep together. Let these moments be the intervals

$$[c_1, d_1], [c_2, d_2], \ldots, [c_{10}, d_{10}]$$

respectively, where $c_1 \leqslant c_2 \leqslant \cdots \leqslant c_{10}$. Note that 2 mathematicians may start to sleep simultaneously when at least 1 of them started to fall asleep. Thus, each c_i is equal to one of the a_j's.

Firstly, since there are 2 mathematicians sleeping at time $t = c_1$, we must have $a_2 \leqslant c_1$. Thus, each of $c_1, c_2, \ldots, c_{10}$ is one of $a_2, a_3, \ldots, a_{10}$. By the pigeonhole principle, two of them are equal. In other words, there are 2 distinct pairs of mathematicians started to fall asleep at the same time. This yields at least 3 mathematicians falling asleep at the same time since the pairs are different.

26. For simplicity, we use sets A, B, C, D, E, F to denote the countries and let k belong to a set if the member with number k comes from that country.

    By the pigeonhole principle, there is a set with $\left\lceil \frac{1978}{6} \right\rceil = 330$ numbers. WLOG assume $a_1 > a_2 > \cdots > a_{330}$ are numbers in A. If one of the numbers

    $$a_1 - a_2,\ a_1 - a_3,\ \ldots,\ a_1 - a_{330}$$

    belongs to A, then we are done. Otherwise, these 329 numbers belong to the other 5 sets. By the pigeonhole principle, there are $\left\lceil \frac{329}{5} \right\rceil = 66$ numbers from the same set. WLOG assume $b_1 > b_2 > \cdots > b_{66}$ belong to B. If one of the numbers

    $$b_1 - b_2,\ b_1 - b_3,\ \ldots,\ b_1 - b_{66}$$

    belongs to B, then we are done. Suppose $b_1 - b_i \in A$. Note that we can write $b_1 = a_1 - a_j$ and $b_i = a_1 - a_k$ for some $j \neq k$. Thus, $b_1 - b_i = a_k - a_j$. We are done in this case as well. For the remaining case, the above 65 numbers belong to the other 4 sets.

    Similarly, by the pigeonhole principle, there are $\left\lceil \frac{65}{4} \right\rceil = 17$ numbers from the same set. WLOG assume $c_1 > c_2 > \cdots > c_{17}$ belong to C. If one of the numbers

    $$c_1 - c_2,\ c_1 - c_3,\ \ldots,\ c_1 - c_{17}$$

    belongs to A, B, C, then we are done as above. Otherwise, by the pigeonhole principle, there are $\left\lceil \frac{16}{3} \right\rceil = 6$ numbers from the same set. WLOG assume $d_1 > d_2 > \cdots > d_6$ belong to D. If one of the numbers

    $$d_1 - d_2,\ d_1 - d_3,\ \ldots,\ d_1 - d_6$$

belongs to A, B, C, D, then we are done. Otherwise, by the pigeonhole principle, there are $\left\lceil \frac{5}{2} \right\rceil = 3$ numbers from the same set. WLOG assume $e_1 > e_2 > e_3$ belong to E. If one of the numbers

$$e_1 - e_2,\ e_1 - e_3$$

belongs to A, B, C, D, E, then we are done. Otherwise, they belong to F. But then the number $(e_1 - e_3) - (e_1 - e_2)$ must belong to one of A, B, C, D, E, F, and we are done in any case as above.

Proof by Contradiction

27. Suppose on the contrary that the integers are all distinct. Then the sum of all integers is at least

$$1 + 2 + \cdots + 2n = \frac{2n(2n+1)}{2} = n(2n+1) > 2n^2.$$

This is a contradiction since it is given that the sum does not exceed $2n^2$. So there are 2 cards having the same integer written on them.

28. Suppose on the contrary that no such triple of vertices exists. Let the vertices be $V_1, V_2, \ldots, V_n$ in that order. WLOG assume V_1 is red and V_2 is blue. Then V_3 cannot be green since otherwise a contradiction arises. Also, V_3 cannot be blue from the condition. So V_3 is red. Inductively, we find that V_{2k-1} is red and V_{2k} is blue for $k \geqslant 1$. As n is odd, V_n is red in colour. This is a contradiction since V_1 and V_n should have distinct colours. So there must be 3 consecutive vertices whose colours are pairwise distinct.

29. We claim that the answer is any finite set of negative real numbers together with the number 0.

Firstly, suppose there exists a positive real number $a \in S$. From the condition, we have $2a = a + a \in S$. Since $2a > 0$, we have $4a = 2a + 2a \in S$. Inductively, we find that $2^k a \in S$ for any positive integer k. Since $a > 0$, all such numbers are distinct. This contradicts the assumption that S is finite.

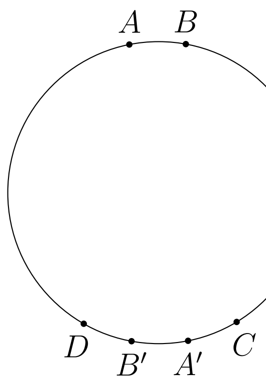
So S contains only non-positive numbers. The set $S = \{0\}$ is clearly an answer. If S contains a negative number a, then $0 = a - a = a + |a| \in S$. So $0 \in S$. Now, for any finite set S of non-positive real numbers containing 0, any $a \in S$ must satisfy $a + |a| = a - a = 0 \in S$. Hence, the condition holds.

30. The people with green hats divide the other people into k blocks of consecutive people whose hats are red or blue for some positive integer k. Suppose on the contrary that each person is a neighbour of at most 1 person with a green hat. Then each block consists of at least 2 people. Thus, the $2k$ people at the end of the k blocks are all different, and each of them is a neighbour of a person with a green hat. Also, they are

the only such people. This yields $2k = 20 + 25 = 45$, which is a contradiction. So there must be a person whose two neighbours both have green hats.

31. There are $5 \times 2 = 10$ ways to divide the chessboard into two. Suppose on the contrary that each of the 10 lines cuts a domino. Since there are $\dfrac{6 \times 6}{2} = 18$ dominoes and each domino is cut by exactly one line, there exists a line cutting at most (and hence exactly) $\left\lfloor \dfrac{18}{10} \right\rfloor = 1$ domino by the pigeonhole principle. This line separates the chessboard into two halves, one with $6a$ cells and the other with $6(6-a)$ cells for some $1 \leqslant a \leqslant 5$. If we remove the domino cut by the line, the remaining $6a - 1$ cells in the first half of the chessboard are tiled completely by some dominoes. This cannot be true as the number of cells is odd but each domino covers 2 cells. Thus, we can always find a suitable division of the chessboard.

32. Suppose on the contrary that there is no such pair of points. Consider any 2 points A and B such that the arc AB has length 1. Let A' and B' be the points on the circle such that AA' and BB' are diameters. From our assumption, A' and B' are not endpoints of any arcs. Therefore, there must be an arc CD of length 3 such that A' and B' lie strictly inside the arc CD. This gives a correspondence between the arcs of length 1 and the arcs of length 3.



Consider the arc DA of the circle which does not contain B and C. Its length is $\dfrac{(1+2+3)n - 4}{2} = 3n - 2$. Suppose there are k arcs of length 1. Then the other $n - 1 - k$ arcs of length 1 lie on another half of the circle. Thus, there are $n - 1 - k$ arcs of length 3 in the arc DA. If there are m arcs of length 2, then the length of this arc is

$$k + 3(n - 1 - k) + 2m = 3n - 3 - 2k + 2m.$$

The parity is different from $3n - 2$, which is a contradiction. Thus, there must be 2 points which are the endpoints of a diameter.

33. Suppose on the contrary that for any partition of the coins into at most n groups, there exists a group with value at least 1. Since there are only finitely many ways to partition the coins, we can choose a partition such that the value of the group with the greatest value is minimized. In case there is more than 1 such partition, we choose any of them for which the number of groups with the greatest value is minimized. Let the values of groups $1, 2, \ldots, n$ be $a_1, a_2, \ldots, a_n$ respectively in such a partition. WLOG assume $a_1 \leqslant a_2 \leqslant \cdots \leqslant a_n$. Since $a_1 + a_2 + \cdots + a_n < \dfrac{n+1}{2}$ and $a_n \geqslant 1$, we have

$$a_1 < \frac{1}{n-1}\Big(\frac{n+1}{2} - 1\Big) = \frac{1}{2}.$$

So we can pick an arbitrary coin in group n and put it in group 1 while keeping the total value of group 1 to be less than 1. In this way, the total value of group n is reduced. So either the value of the group with the greatest value is reduced or the number of groups with the greatest value is reduced. This contradicts the choice of the partition. Therefore, we can always meet the goal.

34. We first introduce some notations for convenience. We use the sets $A_1, A_2, \ldots, A_5$ to denote the keys. Each set contains the labels $1, 2, \ldots$ of the lamps to which the key is connected. The operation $A_i \Delta A_j$ gives the set containing those elements in exactly one of A_i, A_j. Thus, $A_i \Delta A_j$ represents the outcome after the ith and the jth keys are switched. The operation Δ is clearly associative. For simplicity, we use A_{ij} to denote $A_i \Delta A_j$ and A_{ijk} to denote $A_i \Delta A_j \Delta A_k$, etc.

    Suppose on the contrary that $|A_{ijk}| \leqslant 1$ for any indices i, j, k. Consider A_{12}. If $|A_{12}| \geqslant 3$, then since $|A_{134}| \leqslant 1$, there are 2 elements in A_{12} which do not belong to A_{134}. Thus, $|A_{12} \Delta A_{134}| \geqslant 2$. Noting that $A_{234} = A_{12} \Delta A_{134}$, this yields a contradiction.

    If $|A_{12}| \leqslant 2$, then $|A_{12}| = 1$ or 2 since $A_1 \neq A_2$. Note that none of $A_{134}, A_{135}, A_{145}$ may be a singleton set containing an element not in A_{12}. Indeed, if A_{134} is such a set, say, then $A_{234} = A_{12} \Delta A_{134}$ has at least 2 elements, which is a contradiction. Thus, we may assume each of $A_{134}, A_{135}, A_{145}$ is a subset of A_{12}. Also, these three subsets are pairwise distinct, as $A_{134} = A_{135}$ implies $A_4 = A_5$, etc. Therefore, up to permutation, we must have $A_{12} = \{a, b\}$, $A_{134} = \{a\}$, $A_{135} = \{b\}$ and $A_{145} = \varnothing$. Now, we find that

$$A_2 = A_{12} \Delta A_{134} \Delta A_{135} \Delta A_{145} = \varnothing.$$

    This is again a contradiction. Therefore, we can always switch exactly 3 keys to turn at least 2 lamps on.

35. (a) We claim that there is no silver matrix for odd $n > 1$. Suppose on the contrary that such a matrix exists. Suppose the ith row and the ith column form the ith group. Note that each group consists of $2n - 1$ distinct entries in total. So these entries must be $1, 2, \ldots, 2n - 1$ in some order. Thus, each of these numbers appears in all the n groups.

Every entry in the main diagonal belongs to 1 group, while every other entry belongs to 2 groups. Therefore, each of the numbers $1, 2, \ldots, 2n-1$ appears at least $\left\lceil \frac{n}{2} \right\rceil = \frac{n+1}{2}$ times in the matrix. It follows that the matrix consists of at least $\frac{(2n-1)(n+1)}{2}$ entries. This implies $\frac{(2n-1)(n+1)}{2} \leqslant n^2$, i.e. $n \leqslant 1$. This is a contradiction. Hence, there is no silver matrix for odd $n > 1$, in particular for $n = 1997$.

(b) We provide a construction for the case $n = 2^k$ recursively. For $n = 2$, we take $\begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$. Suppose $A = (a_{ij})$ is an $n \times n$ silver matrix. We construct a $2n \times 2n$ silver matrix $B = (b_{ij})$ as follows. We choose $B = \begin{pmatrix} A & A + 2nJ \\ A' + 2nJ & A \end{pmatrix}$. Here J is the matrix whose every entry is 1. Thus, $A + 2nJ$ means we add every entry of A by $2n$. The matrix A' is obtained from A by replacing every entry in the main diagonal by 0. For example, for $n = 4$, we obtain

$$B = \begin{pmatrix} 1 & 2 & 5 & 6 \\ 3 & 1 & 7 & 5 \\ 4 & 6 & 1 & 2 \\ 7 & 4 & 3 & 1 \end{pmatrix},$$

while for $n = 8$, we obtain

$$B = \begin{pmatrix} 1 & 2 & 5 & 6 & 9 & 10 & 13 & 14 \\ 3 & 1 & 7 & 5 & 11 & 9 & 15 & 13 \\ 4 & 6 & 1 & 2 & 12 & 14 & 9 & 10 \\ 7 & 4 & 3 & 1 & 15 & 12 & 11 & 9 \\ 8 & 10 & 13 & 14 & 1 & 2 & 5 & 6 \\ 11 & 8 & 15 & 13 & 3 & 1 & 7 & 5 \\ 12 & 14 & 8 & 10 & 4 & 6 & 1 & 2 \\ 15 & 12 & 11 & 8 & 7 & 4 & 3 & 1 \end{pmatrix}.$$

Now we prove that B is a silver matrix. From the construction, it can be proved easily by induction that B is symmetric in the diagonal from top right to bottom left. So it suffices to check the ith group contains numbers from 1 to $4n - 1$ (since we are working on the case $2n$) for $1 \leqslant i \leqslant n$.

Firstly, $b_{i1}, b_{i2}, \ldots, b_{in}, b_{1i}, b_{2i}, \ldots, b_{ni}$ are the numbers from 1 to $2n - 1$ since they form a group in the smaller matrix A.

Secondly, $b_{i(n+1)}, b_{i(n+2)}, \ldots, b_{i(2n)}, b_{(n+1)i}, b_{(n+2)i}, \ldots, b_{(2n)i}$ excluding $b_{(n+i)i}$ are the numbers from $2n + 1$ to $4n - 1$ since they form a group in the matrix $A + 2nJ$. Together with the entry $b_{(n+i)i} = 2n$, the claim follows. So we can find a silver matrix for the case $n = 2^k$ for any $k \geqslant 1$.